Archit Patil

(469)-325-8093 | architpatil@utexas.edu | Austin, TX

WORK EXPERIENCE

ICAHN SCHOOL OF MEDICINE AT MOUNT SINAI |
Undergraduate Research Intern | Professor Vikas Pejaver
November 2024 – Present | Austin, TX
- Processed 1M+ clinical notes to build a structured dataset mapping rare diseases for downstream analysis.
- Achieved 95% F1-score identifying patients with rare diseases from MIMIC database with BERT-based pipeline.

UNIVERSITY OF TEXAS AT AUSTIN | Undergraduate Research Assistant | Professor David Soloveichik
November 2024 – Present | Austin, TX
- Developed and compared 3 novel Hilbert basis–based algorithms to optimize molecular programming framework, achieving a 25% increase in computational efficiency.

PROJECTS

POSTPARTUM HEMORRHAGE (PPH) CUP - PATENTED
Nov 2021 – Mar 2024 | Maharashtra, India
- Designed a patented PPH cup that quantifies maternal blood loss in real time, lowering estimation errors from 100% to less than 5%.
- Deployed the device in 39 hospitals across India, saving over 100 maternal lives through timely intervention.
- Cut maternal mortality by 50% in critical care by enabling precise blood loss measurement and rapid response.

GAME BOY EMULATOR (C++)
Apr 2025 | Austin, TX
- Built a pixel-accurate Game Boy emulator (SM83 CPU) in C++ using SDL2, implemented full instruction set, memory-mapped I/O, sprite/background rendering, and input handling.

PIPELINED RISC PROCESSOR (VERILOG)
Mar 2025 | Austin, TX
- Designed and programmed a 5-stage pipelined 16-bit RISC processor in Verilog; resolved data/control hazards and integrated instruction forwarding to reduce CPI.

COROUTINE SCHEDULER (C)
Feb 2025 | Austin, TX
- Enabled 1,000+ concurrent tasks via a CSP coroutine framework, reducing context-switch overhead by 40%.

AWARDS

- 2024
  Secured **All India Rank 53** in **JEE Mains** and **Rank 160** in **JEE Advanced** among **1.2 million** candidates (IIT Entrance Examination)
- 2023–2024
  Ranked top 1% nationally in India in Physics, Astronomy, and Chemistry Olympiads

EDUCATION

UNIVERSITY OF TEXAS AT AUSTIN
Bachelor of Science in Computer Science (Turing scholar honors), Math and Computational Biology
Expected May 2027 | Austin, TX
College of Natural Sciences
Cum. GPA: 3.732 / 4.0

SKILLS

PROGRAMMING
1+ years:
Python • Java • C • C++
HTML • JavaScript • SQL
LaTeX • Git
PyTorch • TensorFlow • Assembly (ARM, x86) • Verilog

TECHNOLOGY
Git/Github • Data Structures • Deep Learning
Cryptography

COURSEWORK

UNDERGRADUATE
Data Structures
Discrete Math
Number Theory
Calculus I-III
Probability I
Real Analysis
Differential Equations and Linear Algebra
Linear Algebra
Computer Architecture
Intro to Microeconomics
Operating Systems (In progress)

CLUBS

Directed Reading Program (DiRP)
ECLAIR Robotics Association for Computing Machinery (ACM)w

LINKS

Github:// **ArchitRahul**
LinkedIn:// **Archit Patil**